Unified prefetch-chain registration in GTP

Sharding selects weights • Registration selects membership • Runtime manages materialization

1 Apply sharding policy

Model weights

↓

Sharding policy
Exclusions + optional opt-ins

↓

GTP-sharded weights
Dense + MoE projections

Optional weights enter only when their sharding is enabled.

Replicated weights
No GTP weight gather

2 Register through one API

Model setup registers remaining weights; explicit overrides win.

`register_prefetch_chain("default_dense", dense_weights)`

Dense weight 0 → **Dense weight 1** → **Dense weight 2**

`register_prefetch_chain("default_expert_fc1", fc1_weights)`

block0/GFC1 (local experts, coalesced AG) → **block1/GFC1** → **block2/GFC1**

`register_prefetch_chain("default_expert_fc2", fc2_weights)`

block0/GFC2 (local experts, coalesced AG) → **block1/GFC2** → **block2/GFC2**

Finalize captured/eager boundaries per parallel group; then warmup learns execution links within each chain.

3 Execute the schedule

Default stream mapping shown below; logical chains do not require stream sharing.

Dense all-gather stream

Expert all-gather stream

AG(GFC1) → AG(GFC2)

Same group + capture mode → shared AG stream

Temporary weight buffers per chain

Dense | GFC1 | GFC2

Consumers wait for their own weights.

Forward: next weight | dgrad: previous weight

Recompute-forward: learn a separate order for recomputed operators within each chain.